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香港考試及評核局
HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

2022年香港中學文憑
HONG KONG DIPLOMA OF SECONDARY EDUCATION 2022

CHEMISTRY PAPER 2

MARKING SCHEME

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INSTRUCTIONS TO MARKERS

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well reasoned.

3. The following symbols are used:

/	A single slash indicates an acceptable alternative within an answer.
*	Step-mark (for questions involving calculations)
†	Correct spelling required

4. In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

5. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

6. Award zero marks for answers which are contradictory.

7. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

Marks

1. (a) (i) (1) A catalyst is used. / HI/Rh is a catalyst. / Methanol can be made from renewable / biomass. / Atom economy is 100% / (very) high. / CH_3COOH is the only product. / No waste is produced. 1

(2) Methanol / CO(g) is toxic. / HI (or CH_3COOH) is corrosive. 1

(ii) (1) It can increase the surface area of the catalyst so as to increase the effectiveness of the catalyst. 1

(2) Catalysts can be poisoned by impurities / lead compounds. / The active sites of catalysts are blocked. 1

(iii) glass bottle 1

(b) (i) water / H_2O 1

(ii) (1) chlorine / Cl_2 1

(2) The concentration of $Cl^-(aq)$ ions is much higher than that of $OH^-(aq)$ ions, $Cl^-(aq)$ ions are preferentially discharged. (A comparative sense) 1

(iii) (1) $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ / $2H^+(aq) + 2e^- \rightarrow H_2(g)$ (State symbols not required)(Ignore incorrect state symbols) 1

(2)
- $OH^-(aq)$ ions are continuously formed in the cathode part. / $H^+(aq)$ ions are discharged while $OH^-(aq)$ ions are remained in the cathode part. 1
- The ion-permeable membrane only allows $Na^+(aq)$ ions but not $Cl^-(aq)$ ions to pass to the cathode part. 1

(iv) sodium hypochlorite / NaOCl / NaClO /sodium chlorate / $NaClO_3$ 1

Marks

1. (c) (i) 'Initial rate' is the (instantaneous) rate at the start of a reaction / the rate at time zero. 1

(ii)
- Since $[H^+(aq)]$ is much higher than $[S_2O_3^{2-}(aq)]$, $[H^+(aq)]$ changes only a little / the change in $[H^+(aq)]$ is negligible at the start of the reaction. 1
- $[H^+(aq)]$ is regarded as constant, / then $k[H^+(aq)]^b$ can be regarded as a constant / only $[S_2O_3^{2-}(aq)]$ is a variable. 1

(iii) rate = $k'[S_2O_3^{2-}(aq)]^a$
log (rate) = *log* k' + *a log*($[S_2O_3^{2-}(aq)]$) (*a* = the slope of the straight line) 1*
Slope = [–1.10 – (–1.50)] ÷ [–1.84 – (–2.24)] (If not use the dotted lines, 0 mark) 1*
= 1
It is first order with respect to $S_2O_3^{2-}(aq)$. 1
(1 mark for writing correct log equation,
1 mark for substituting correct data,
1 mark for giving correct answer)
(Accept : other methods of calculations)

(iv) $log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.3R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$log\,(1.9) = \frac{E_a}{2.3\times 8.31}\left(\frac{1}{298} - \frac{1}{308}\right)$ 1*

Ea = (+)48.90 kJ mol^{-1} 1
(Accept: 48.805 – 49.004;
Accept : no unit ;
Not accept: unit in kJ)

Marks

2. (a) (i) (1) Particles of materials with sizes between 1–100 nm / less than 100 nm 1

(2) making stained glass / colour windows 1

(ii) (1) Diagram **C** 1

(2) nematic phase 1

(iii) Atom economy = $\frac{32.0}{2\times40.0+71.0+2\times17.0} \times 100\% = 17.3\%$ (Accept: 17%) 1

(b) (i) face-centred cubic / cubic close-packed † 1

(ii) No. of aluminium atoms in the unit cell = $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$ (need to show the step) 1

(iii) Density of aluminium = $\underline{27.0 \times 4} \div (6.02 \times 10^{23}) \underline{\div (4.05 \times 10^{-8})^3}$ 1*

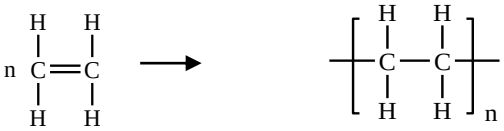
$= \underline{2.70}$ g cm^{-3} 1

(Accept: 2.7 g dm^{-3}; no need to show the unit)

(iv) (1)
- In the duralumin sample, atoms of different sizes added (Cu/Mg/Mn atoms) distort the regular packing of atoms in pure aluminium / become irregular packing / become not closely packed. 1
- This reduces the (strength of) metallic bonding and leads to a lower melting point. 1

(2) Agree.
This kind of duralumin has a greater tensile strength /stronger than pure aluminium. 1
(Accept: greater hardness / rigidity / toughness)
(Do not accept: lower density / higher corrosion resistance)

Or, Not agree.
This kind of duralumin has a lower melting point. (1)

(c) (i)  1

(ii) addition (polymerisation) † 1

(iii) (1)
- HDPE has a linear structure / linear polymer chains / with less branches, the chains are packed more closely. 1
- There are stronger van der Waals' forces between polymer chains in HDPE than in LDPE. 1

(2) blow moulding 1

(iv) (1) thermosetting / low thermal conductivities (good heat insulation) 1

(2) Because there are strong cross-links / covalent bonds between polymer chains. 1

(3) resistant to corrosion / flexible / bent easily 1
(Do not accept thermal properties related answers)

Marks

3. (a) (i)
- Pass the two gases to $K_2Cr_2O_7/H^+(aq)$ separately. 1
- $SO_2(g)$ can turn $K_2Cr_2O_7/H^+(aq)$ from orange to green, while there is no observable change for $CO_2(g)$. 1
  (A comparative sense)
  (If H^+ is not present in $K_2Cr_2O_7$, 0 mark for both reagent and observation.)

Or, Using $KMnO_4/H^+(aq)$,
$SO_2(g)$ can turn $KMnO_4/H^+(aq)$ from purple to colourless / (very) pale pink, while there is no observable change for $CO_2(g)$.
(If H^+ is not present in $KMnO_4$, 0 mark for both reagent and observation.)

Or, Using $Br_2(aq)$ or Br_2(in organic solvent),
$SO_2(g)$ can turn $Br_2(aq)$ from reddish brown to colourless / $SO_2(g)$ can turn Br_2(in organic solvent) from orange/brown to colourless, while there is no observable change for $CO_2(g)$.
(Not accept: red or yellow for colour of $Br_2(aq)$ or Br_2(in organic solvent))
(Not accept: $Br_2(l)/Br_2(g)$ as the reagent, BUT can give mark to correct observation)

(Accept: other possible chemical tests)

(ii) The mass spectra of CH_3CH_2CHO and CH_3COCH_3 are recorded respectively.
- In the mass spectrum of CH_3CH_2CHO, a significant peak appears at m/z = 29 / corresponding to CHO^+ / $CH_3CH_2^+$, while this peak does not appear in the mass spectrum of CH_3COCH_3. 1
- In the mass spectrum of CH_3CH_2CHO, a significant peak appears at m/z = 57 / corresponding to $CH_3CH_2CO^+$, while this peak does not appear in the mass spectrum of CH_3COCH_3. 1

Or, In the mass spectrum of CH_3COCH_3, a significant peak appears at m/z = 43 / corresponding to CH_3CO^+, while this peak does not appear in the mass spectrum of CH_3CH_2CHO. (1)

(A comparative sense)

(iii) (Anhydrous) sodium sulphate / Na_2SO_4 1

(b) (i) The maximum mass of **Y** can dissolve in 50 cm^3 water at 80°C = 3.04 × 50/100= 1.52 g. 1
As the mass of **Y** in the solid sample should be less than 1.4 g, therefore all **Y** should have dissolved.
(or 3.04 g to compare with 2.8 g)

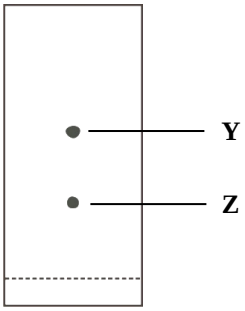
(ii) To remove the water-insoluble activated charcoal. 1

(iii) crystallisation † 1

(iv) Some **Y** do not crystallise / are left on the filter paper / are washed away. 1
(Not accept: 'Loss during the steps', need to mention the specific step.)

Marks

3. (b) (v) (1)



2

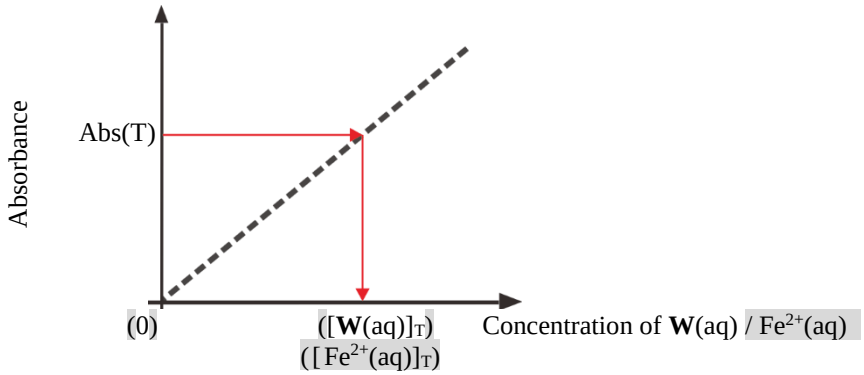
(1 mark for sketching correct chromatogram (2 points and 1 line for starting position / 3 points with one point representing the starting position);
1 mark for correct labels (**Y** and **Z**; **Y** is higher than **Z**))

(2) **Y**,
because **Y** has a larger R_f value / **Y** moves faster / **Y** takes a shorter time to reach the bottom of the column. 1

(c) (i) (1)
- pale yellow colour: $Fe^{3+}(aq)$ ions 1
- pale pink colour: $MnO_4^-(aq)$ ions 1

(2) concentration of $Fe^{2+}(aq)$ ions in solution **S** = $0.0041 \times 32.35 \div 25 \times 5$ 1*
$= 0.0265$ M 1
(Accept: 0.027, 0.02653)

(ii) (1) 3



(1 mark for straight line of calibration curve;
1 mark for labels of axes;
1 mark for two lines to find [**W**(aq)]$_T$)

(2) Concentration of $Fe^{2+}(aq)$ ion in solution **S**
= Concentration of **W**(aq) in solution **T** $\times$ 100 1
= ([**W**(aq)]$_T \times 100$) or ([Fe^{2+}(aq)]$_T \times 100$)